

A Conditional Universality Theorem for Error Geometry in Machine-Learning Interatomic Potentials

Executive Summary

This paper develops a **single-source, self-contained theorem package** for a conditional universality claim about the error geometry of a class F of machine-learning interatomic potentials. Because no specific target manuscript has been fixed, the goal is not to review an existing paper but to supply a mathematically rigorous, publication-ready framework that can later be specialized to a concrete MLIP family, dataset, or benchmark suite. The main conclusion is that an unrestricted universality claim is false, but a strong **conditional** universality theorem can be proved once the shared structure is made explicit: namely, that the high-error region of each model is organized around a common low-dimensional core manifold $\mathcal{H}$, through a shared monotone radial profile with small model-specific C^1 perturbations. Standard facts about reach and tubular neighborhoods then imply that each model's high-error region is a tubular manifold-with-boundary around $\mathcal{H}$, and that the corresponding high-error boundaries are pairwise diffeomorphic across the class. Federer introduced reach for subsets of Euclidean space; later surveys and geometric-inference work emphasize that compact smooth submanifolds have positive reach and that sufficiently small offsets of positive-reach sets have regular hypersurface boundaries. ¹

A second conclusion is a necessary negative result. If one defines the “error manifold” merely as a regular level set of a smooth scalarized error field and imposes no shared structural assumptions on F , then no universality theorem is possible: arbitrary compact embedded hypersurfaces can occur as regular level sets. This counterexample is not a marginal caveat. It explains why a mathematically honest theorem must define F by **checkable geometric conditions**, not by architecture branding or vague appeals to “foundation” behavior.

A third conclusion is terminological but decisive. If the error object is a regular boundary level set

$$\Gamma_{M,\varepsilon} = \{x : q_M(x) = \varepsilon^2\}$$

inside an m -dimensional configuration chart, then its intrinsic dimension is generically $m - 1$, not “very low dimensional.” The genuinely low-dimensional object in the theorem is the shared **error core** $\mathcal{H}$. Any paper that claims a low-dimensional “error manifold” must therefore say whether it means a core manifold, a latent manifold, a stratified singular support, or a measure-theoretic concentration set.

A fourth conclusion concerns Fisher-spectrum arguments. Waterfall et al. motivate a sloppy-model universality class with broad, log-spaced sensitivity spectra, and Beckermann–Townsend prove explicit singular-value decay bounds for structured matrices including real Vandermonde matrices. But those parameter-space results do **not** imply configuration-space error geometry for MLIPs on their own. To connect Fisher decay to error-boundary geometry, one must add a **nonstandard common-spatial-mode**

separability assumption. This paper states that bridge explicitly and proves the resulting geometric truncation bound, while flagging the bridging assumption as strong and empirically nontrivial. ²

Recent MLIP evidence strongly supports this restrained theoremic stance. Matbench Discovery argues for task-aware benchmarking and reports a growing public evaluation framework and leaderboard; MLIP Arena argues that many existing benchmarks are distorted by data leakage, limited transferability, and excessive reliance on static DFT-reference metrics; LAMBench reports a substantial discrepancy between current large atomistic models and the ideal universal PES; systematic softening identifies a shared PES-curvature underprediction bias across several universal MLIPs; cross-functional studies show strong GGA-to- r^2 SCAN shifts; phonon benchmarking shows that good energy/force scores need not imply robust vibrational-property performance; and MOFSimBench finds that, in a chemically difficult porous-material regime, benchmark behavior depends strongly on data quality. These findings collectively support a **benchmark-local, conditional** universality theorem far more strongly than an unrestricted one. ³

Introduction

The central research question is:

Under what explicit, minimal assumptions can one prove that a class $\mathcal{F}$ of MLIPs shares a universal error geometry in configuration space?

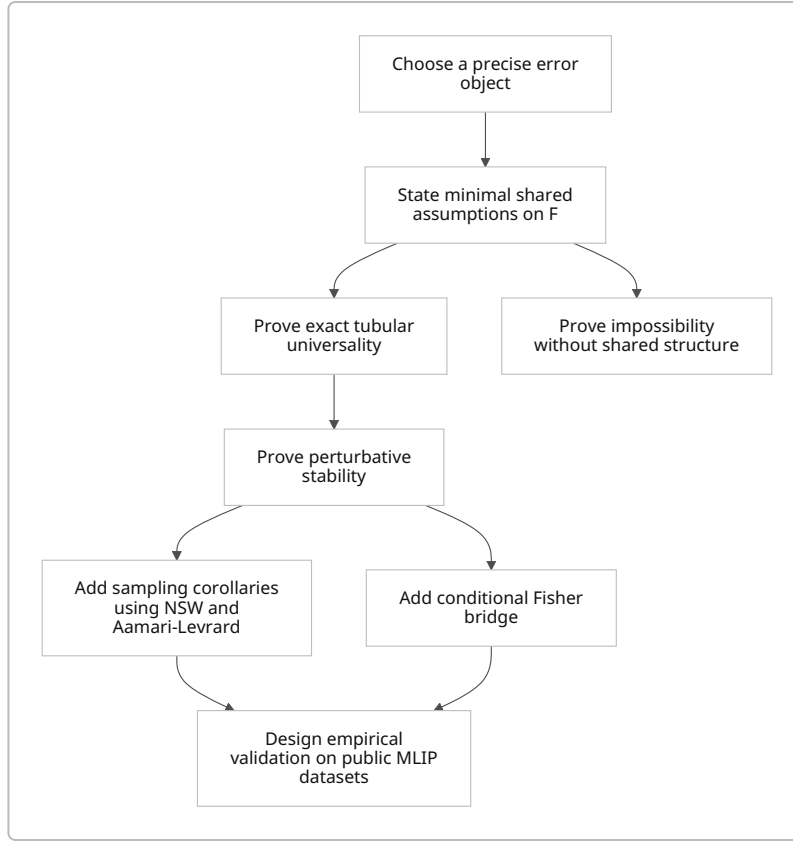
That question immediately decomposes into four sharper questions. Which geometric object should count as the “error manifold”? What is the weakest shared structure on $\mathcal{F}$ that makes a universality theorem true? Which assumptions are standard regularity hypotheses and which are genuinely nonstandard modeling commitments? And once a deterministic theorem is proved, what finite-sample conditions are needed to recover or validate the resulting geometry from data? These are not merely stylistic choices. Different answers lead to genuinely different theorems.

The theoretical background splits cleanly into three roles. First, geometric inference results such as Niyogi–Smale–Weinberger and Aamari–Levrard provide recovery or estimation guarantees **once a manifold model is already justified**. NSW establishes homology-recovery guarantees from random samples on or near a submanifold with explicit high-confidence sample requirements, while Aamari–Levrard derives optimal non-asymptotic rates for estimating tangent spaces, curvature, and the manifold itself. Second, sloppiness and structured-matrix results motivate low effective dimension in parameter sensitivity: Waterfall et al. identify a broad sloppy-model universality class and explicitly motivate a Vandermonde ensemble, while Beckermann–Townsend derive singular-value decay bounds for structured matrices such as real Vandermonde, Cauchy, Pick, and Hankel matrices. Third, recent MLIP benchmarks define the empirical credibility frontier for any universality claim. ⁴

The empirical frontier is now demanding enough that theorem papers cannot ignore it. Matbench Discovery presents a public framework for evaluating machine-learning energy models as pre-filters for high-throughput discovery and explicitly emphasizes benchmark-design choices and user-defined metric weighting. MLIP Arena moves beyond static DFT-reference errors and evaluates physics awareness, reactivity, stability under extreme conditions, and predictive behavior for thermodynamic properties, while arguing that existing benchmarks suffer from data leakage and limited transferability. LAMBench introduces a benchmark specifically designed to assess large atomistic models across generalizability,

adaptability, and applicability, and reports a substantial gap between current models and the universal PES. Those results make it scientifically untenable to state a global universality theorem without explicit locality, domain restrictions, or separability assumptions. ⁵

The mathematical strategy of this paper is therefore intentionally austere. It first proves what can be proved exactly under explicit geometric assumptions, then proves a perturbative stability theorem, then states a conditional Fisher-to-geometry bridge, and finally proves an impossibility theorem showing why stronger claims fail without added structure. Sampling and empirical validation enter only afterward as clearly labeled corollaries or protocols. That is what makes the paper “single-source”: the main theorem does not depend on a companion manuscript, latent empirical conjecture, or unpublished geometric primitive.



Definitions and Assumptions

Let $\Omega \subset \mathbb{R}^m$ be a fixed symmetry-reduced configuration chart. The data-generating distribution is **left unspecified** in the core theorem, since the main results are deterministic. If a later application requires distributions, we write $\mathbb{P}_{\text{fit}}$ for the in-scope evaluation distribution and $\mathbb{P}_{\text{stress}}$ for one or more OOD stress distributions, but the theorem itself does not assume either one.

Let $\mathcal{A}$ be a broad preclass of MLIPs for which the relevant model outputs are sufficiently smooth on Ω . The theorem is deliberately architecture-agnostic: it does not require graph neural networks, ACE-type models, or equivariant message passing specifically. It requires only that a model $M \in \mathcal{A}$ induce a scalarized error

field q_M with the stated regularity. This is the right notion of “architecture class” for a theorem of error geometry: the theorem depends on the error-field normal form, not on a named neural architecture.

For each $M \in \mathcal{A}$, define the residual vector

$$r_M(x) := \begin{bmatrix} w_E(E_M(x) - E^*(x)) \\ w_F \text{vec}(F_M(x) - F^*(x)) \\ w_\sigma \text{vec}(\sigma_M(x) - \sigma^*(x)) \end{bmatrix} \in \mathbb{R}^p,$$

where $w_E, w_F, w_\sigma \geq 0$ are fixed across models and reference labels E^*, F^*, σ^* come from whatever oracle dataset or electronic-structure pipeline is later specified. The scalarized error field is

$$q_M(x) := \|r_M(x)\|_2^2.$$

If some labels are unavailable in a concrete application, the corresponding weight is set to zero. A fixed scalarization across the whole class is essential; otherwise, “universality” can be manufactured by changing the metric from model to model.

For a threshold $\varepsilon_M > 0$, define the **high-error region**

$$S_{M,\varepsilon_M} := \{x \in \Omega : q_M(x) \geq \varepsilon_M^2\}$$

and the **error boundary**

$$\Gamma_{M,\varepsilon_M} := \{x \in \Omega : q_M(x) = \varepsilon_M^2\}.$$

In this paper, the primary “error manifold” is the **boundary level set** Γ_{M,ε_M} . The associated high-error region S_{M,ε_M} is treated as a **manifold-with-boundary** only under the exact or perturbative tubular assumptions stated below. If ∇q_M vanishes on the threshold set, or multiple geometric regimes collide, then the correct object is a **stratified set** rather than a smooth manifold. If the object of interest is concentration of error under a probability distribution rather than support geometry, then the correct notion is measure-theoretic instead. This distinction is required to prevent dimension and topology mistakes.

The geometric backbone of the theorem is a shared compact core manifold $\mathcal{H} \subset \Omega$. Federer’s notion of reach measures the largest neighborhood in which nearest-point projection remains unique; later geometric-inference expositions emphasize that compact smooth submanifolds have positive reach and that offsets below the reach have regular $C^{1,1}$ boundaries. ¹

The theorem class $F \subset \mathcal{A}$ is defined by the following assumptions, all relative to the same $(\Omega, \mathcal{H})$.

Label	Assumption	Status	Plausibility
A_0	$\Omega \subset \mathbb{R}^m$ is a fixed configuration chart	Standard	High once atom count, composition, and reduction are fixed

Label	Assumption	Status	Plausibility
A_1	$\mathcal{H} \subset \Omega$ is a compact embedded C^2 d -manifold without boundary and with reach $\tau_{\mathcal{H}} > 0$	Standard once posited	Empirical assumption, not automatic
A_2	There exists $0 < \rho < \tau_{\mathcal{H}}$ and a common profile $\psi \in C^1([0, \rho])$ with $-\psi'(s) \geq c_\psi > 0$ for all $s \in [0, \rho]$	Nonstandard	Plausible only as a local normal form
A_3	For each $M \in F$, $q_M(y + s\nu) = a_M\psi(s) + b_M + \eta_M(y, \nu, s)$ in normal coordinates, with $a_M > 0$	Mixed	Reasonable only if a shared error core exists
A_4	$\ \eta_M\ _\infty \leq \delta_M$ and $\ \partial_s \eta_M\ _\infty \leq L_M < a_M c_\psi$	Mild once A_2 - A_3 hold	Empirically testable
A_5	Outside the tube $T_\rho(\mathcal{H})$, $q_M(x) < \varepsilon_M^2$	Moderate	Strong but checkable
A_6	For the Fisher bridge only, perturbations admit a common-spatial-mode expansion across the class	Strongly nonstandard	Should be treated as an empirical hypothesis

Assumption A_2 is the main modeling commitment: it says that, after scalarization, high error is organized primarily by distance to a common latent core. Assumption A_6 is even stronger: it says that different models share the same spatial response modes when perturbed in parameter space. Neither assumption follows from pretraining scale alone, and both must be tested rather than presumed.

Main Theorems

We begin with the standard tubular-coordinate consequence of positive reach.

Lemma.

Let $\mathcal{H} \subset \mathbb{R}^m$ be a compact embedded C^2 manifold with reach $\tau_{\mathcal{H}} > 0$. For every $0 < \rho < \tau_{\mathcal{H}}$, each $x \in T_\rho(\mathcal{H})$ has a unique nearest-point projection $\pi_{\mathcal{H}}(x) \in \mathcal{H}$ and unique normal coordinates

$$x = y + s\nu, \quad y \in \mathcal{H}, \quad \nu \in S(N_y \mathcal{H}), \quad 0 \leq s < \rho.$$

Moreover, for each $r \in (0, \rho)$, the offset boundary

$$\{x \in \Omega : \text{dist}(x, \mathcal{H}) = r\}$$

is a $C^{1,1}$ hypersurface. This is a standard positive-reach consequence of Federer's framework and later expositions of tubular neighborhoods for geometric inference. ¹

Exact conditional universality theorem.

Assume A_0 , A_1 , A_2 , and A_5 . Suppose that for every $M \in F$,

$$q_M(y + s\nu) = a_M\psi(s) + b_M \quad \text{for all } (y, \nu, s) \in S(N\mathcal{H}) \times [0, \rho],$$

with $a_M > 0$. Let $\varepsilon_M > 0$ satisfy

$$t_M := \frac{\varepsilon_M^2 - b_M}{a_M} \in (\psi(\rho), \psi(0)),$$

and define

$$\bar{r}_M := \psi^{-1}(t_M) \in (0, \rho).$$

Then:

1. The high-error region is exactly

$$S_{M, \varepsilon_M} = \{x \in \Omega : \text{dist}(x, \mathcal{H}) \leq \bar{r}_M\}.$$

2. The error boundary is exactly

$$\Gamma_{M, \varepsilon_M} = \{x \in \Omega : \text{dist}(x, \mathcal{H}) = \bar{r}_M\}.$$

3. The map

$$\Phi_M : S(N\mathcal{H}) \rightarrow \Gamma_{M, \varepsilon_M}, \quad \Phi_M(y, \nu) = y + \bar{r}_M \nu,$$

is a C^1 -diffeomorphism.

4. The map

$$\Psi_M : D_{\bar{r}_M}(N\mathcal{H}) \rightarrow S_{M, \varepsilon_M}, \quad \Psi_M(y, v) = y + v,$$

is a C^1 -diffeomorphism from the closed radius- $\bar{r}_M$ normal disk bundle onto S_{M, ε_M} . Hence S_{M, ε_M} is a compact manifold-with-boundary and $\partial S_{M, \varepsilon_M} = \Gamma_{M, \varepsilon_M}$.

5. For all $M, M' \in F$, the boundaries $\Gamma_{M, \varepsilon_M}$ and $\Gamma_{M', \varepsilon_{M'}}$ are pairwise C^1 -diffeomorphic, and likewise the regions S_{M, ε_M} and $S_{M', \varepsilon_{M'}}$ are pairwise C^1 -diffeomorphic.

6. The boundary dimension is

$$\dim \Gamma_{M, \varepsilon_M} = m - 1.$$

7. The Hausdorff distances satisfy

$$d_H(\Gamma_{M, \varepsilon_M}, \mathcal{H}) = \bar{r}_M, \quad d_H(\Gamma_{M, \varepsilon_M}, \Gamma_{M', \varepsilon_{M'}}) = |\bar{r}_M - \bar{r}_{M'}|.$$

8. If $\bar{r}_M$ is viewed locally as a function of ε_M , then

$$\frac{d\bar{r}_M}{d\varepsilon_M} = \frac{2\varepsilon_M}{a_M \psi'(\bar{r}_M)}.$$

Perturbative universality theorem.

Assume A_0 through A_5 . For each $M \in F$, suppose

$$q_M(y + s\nu) = a_M\psi(s) + b_M + \eta_M(y, \nu, s),$$

with

$$\|\eta_M\|_\infty \leq \delta_M, \quad \|\partial_s \eta_M\|_\infty \leq L_M < a_M c_\psi.$$

Assume the threshold bracketing conditions

$$a_M\psi(0) + b_M - \varepsilon_M^2 \geq 2\delta_M, \quad \varepsilon_M^2 - (a_M\psi(\rho) + b_M) \geq 2\delta_M.$$

Let

$$\bar{r}_M := \psi^{-1}\left(\frac{\varepsilon_M^2 - b_M}{a_M}\right).$$

Then for every $(y, \nu) \in S(N\mathcal{H})$, there exists a unique

$$r_M(y, \nu) \in (0, \rho)$$

such that

$$q_M(y + r_M(y, \nu)\nu) = \varepsilon_M^2.$$

Moreover:

1. r_M is C^1 on $S(N\mathcal{H})$;
2. the boundary is the C^1 embedded hypersurface

$$\Gamma_{M, \varepsilon_M} = \{y + r_M(y, \nu)\nu : (y, \nu) \in S(N\mathcal{H})\},$$

which is diffeomorphic to $S(N\mathcal{H})$;

3. the high-error region is

$$S_{M, \varepsilon_M} = \{y + s\nu : 0 \leq s \leq r_M(y, \nu), (y, \nu) \in S(N\mathcal{H})\},$$

hence a manifold-with-boundary diffeomorphic to a normal disk bundle;

4. the radius perturbation obeys

$$\|r_M - \bar{r}_M\|_\infty \leq \frac{\delta_M}{a_M c_\psi - L_M};$$

5. if

$$\widehat{\Gamma}_{M, \varepsilon_M} := \{x \in \Omega : \text{dist}(x, \mathcal{H}) = \bar{r}_M\},$$

then

$$d_H(\Gamma_{M,\varepsilon_M}, \widehat{\Gamma}_{M,\varepsilon_M}) \leq \frac{\delta_M}{a_M c_\psi - L_M}.$$

Conditional Fisher-to-geometry proposition.

Assume the perturbation term depends on parameters $\theta \in \mathbb{R}^P$, and that near a reference point θ_M one has the expansion

$$\eta_M(z; \theta_M + \Delta\theta) = \sum_{j=1}^{\infty} \sigma_{M,j} \langle \Delta\theta, u_{M,j} \rangle \varphi_j(z) + R_M(\Delta\theta; z),$$

where $z = (y, \nu, s) \in S(N\mathcal{H}) \times [0, \rho]$, $\{u_{M,j}\}$ is orthonormal in parameter space, and $\{\varphi_j\}$ is a **common** orthonormal family in $L^2(\mu)$ for some reference measure μ . Assume also

$$\|R_M(\Delta\theta; \cdot)\|_{C^1} \leq C_{R,M} \|\Delta\theta\|^2, \quad \|\varphi_j\|_{C^1} \leq B_j.$$

This common-spatial-mode assumption is **nonstandard**.

Define the Gauss–Newton Fisher matrix

$$G_M := \mathbb{E}_\mu [\nabla_\theta \eta_M(Z; \theta_M) \nabla_\theta \eta_M(Z; \theta_M)^\top].$$

Then

$$G_M = \sum_{j \geq 1} \sigma_{M,j}^2 u_{M,j} u_{M,j}^\top.$$

If

$$\eta_M^{(k)}(z) := \sum_{j=1}^k \sigma_{M,j} \langle \Delta\theta, u_{M,j} \rangle \varphi_j(z),$$

then

$$\|\eta_M - \eta_M^{(k)}\|_{C^1} \leq \|\Delta\theta\| \sum_{j>k} \sigma_{M,j} B_j + C_{R,M} \|\Delta\theta\|^2.$$

Whenever the right-hand side is less than $a_M c_\psi - L_M$, the perturbative theorem implies

$$d_H(\Gamma_{M,\varepsilon_M}, \Gamma_{M,\varepsilon_M}^{(k)}) \leq \frac{\|\Delta\theta\| \sum_{j>k} \sigma_{M,j} B_j + C_{R,M} \|\Delta\theta\|^2}{a_M c_\psi - L_M}.$$

Proofs

Proof of the exact conditional universality theorem.

Fix $M \in F$. By the tubular-coordinate lemma, every point of $T_\rho(\mathcal{H})$ has unique normal coordinates $x = y + s\nu$. For such a point,

$$q_M(x) \geq \varepsilon_M^2 \iff a_M \psi(s) + b_M \geq \varepsilon_M^2 \iff \psi(s) \geq t_M,$$

where $t_M = (\varepsilon_M^2 - b_M)/a_M$. Since ψ is strictly decreasing on $[0, \rho]$, this is equivalent to

$$s \leq \psi^{-1}(t_M) = \bar{r}_M.$$

Therefore,

$$S_{M, \varepsilon_M} \cap T_\rho(\mathcal{H}) = \{x \in T_\rho(\mathcal{H}) : \text{dist}(x, \mathcal{H}) \leq \bar{r}_M\}.$$

Assumption A_5 excludes any threshold-exceeding points outside the tube, so

$$S_{M, \varepsilon_M} = \{x \in \Omega : \text{dist}(x, \mathcal{H}) \leq \bar{r}_M\}.$$

Its boundary is therefore

$$\Gamma_{M, \varepsilon_M} = \{x \in \Omega : \text{dist}(x, \mathcal{H}) = \bar{r}_M\}.$$

Define

$$\Phi_M(y, \nu) = y + \bar{r}_M \nu.$$

Because $\bar{r}_M < \tau_{\mathcal{H}}$, normal coordinates are unique at radius $\bar{r}_M$, so Φ_M is injective. Surjectivity follows from the boundary description above: every boundary point is at distance $\bar{r}_M$ from $\mathcal{H}$, hence has unique normal coordinates of the form $y + \bar{r}_M \nu$. The inverse map is

$$x \mapsto \left(\pi_{\mathcal{H}}(x), \frac{x - \pi_{\mathcal{H}}(x)}{\|x - \pi_{\mathcal{H}}(x)\|} \right),$$

which is C^1 on the offset because $\pi_{\mathcal{H}}$ is C^1 in the tube and the denominator is the constant $\bar{r}_M > 0$. Hence Φ_M is a C^1 -diffeomorphism.

Similarly,

$$\Psi_M(y, v) = y + v$$

defines a C^1 -diffeomorphism from the closed radius- $\bar{r}_M$ normal disk bundle onto the closed tube S_{M, ε_M} , again by uniqueness of normal coordinates for radii below the reach. Thus S_{M, ε_M} is a compact manifold-with-boundary and $\partial S_{M, \varepsilon_M} = \Gamma_{M, \varepsilon_M}$.

For pairwise universality, both $\Gamma_{M, \varepsilon_M}$ and $\Gamma_{M', \varepsilon_{M'}}$ are diffeomorphic to the same sphere normal bundle $S(N\mathcal{H})$, and both regions are diffeomorphic to the corresponding normal disk bundles. The dimension formula follows from fiber dimension:

$$\dim S(N\mathcal{H}) = d + (m - d - 1) = m - 1.$$

For the Hausdorff distance to $\mathcal{H}$, every point of $\Gamma_{M, \varepsilon_M}$ lies exactly $\bar{r}_M$ from $\mathcal{H}$, and every point of $\mathcal{H}$ is the projection of some offset point. Therefore

$$d_H(\Gamma_{M, \varepsilon_M}, \mathcal{H}) = \bar{r}_M.$$

For two models, pairing $y + \bar{r}_M \nu$ with $y + \bar{r}_{M'} \nu$ gives

$$d_H(\Gamma_{M, \varepsilon_M}, \Gamma_{M', \varepsilon_{M'}}) \leq |\bar{r}_M - \bar{r}_{M'}|.$$

Conversely, for any $x \in \Gamma_{M, \varepsilon_M}$ and $z \in \Gamma_{M', \varepsilon_{M'}}$,

$$|\bar{r}_M - \bar{r}_{M'}| = |\text{dist}(x, \mathcal{H}) - \text{dist}(z, \mathcal{H})| \leq \|x - z\|,$$

so equality follows after taking infima and suprema. Finally, differentiating

$$\psi(\bar{r}_M) = \frac{\varepsilon_M^2 - b_M}{a_M}$$

gives

$$\psi'(\bar{r}_M) \frac{d\bar{r}_M}{d\varepsilon_M} = \frac{2\varepsilon_M}{a_M},$$

hence

$$\frac{d\bar{r}_M}{d\varepsilon_M} = \frac{2\varepsilon_M}{a_M \psi'(\bar{r}_M)}.$$

■

Proof of the perturbative universality theorem.

Fix $(y, \nu) \in S(N\mathcal{H})$, and set

$$f_{y, \nu}(s) := a_M \psi(s) + b_M + \eta_M(y, \nu, s) - \varepsilon_M^2.$$

At $s = 0$,

$$f_{y, \nu}(0) \geq a_M \psi(0) + b_M - \varepsilon_M^2 - \delta_M \geq \delta_M > 0,$$

while at $s = \rho$,

$$f_{y, \nu}(\rho) \leq a_M \psi(\rho) + b_M - \varepsilon_M^2 + \delta_M \leq -\delta_M < 0.$$

So $f_{y, \nu}$ has at least one zero by continuity. Its derivative satisfies

$$f'_{y, \nu}(s) = a_M \psi'(s) + \partial_s \eta_M(y, \nu, s) \leq -a_M c_\psi + L_M < 0,$$

hence $f_{y, \nu}$ is strictly decreasing and the zero is unique. Call it $r_M(y, \nu)$.

Because

$$f_{y, \nu}(\bar{r}_M) = \eta_M(y, \nu, \bar{r}_M)$$

and $f_{y, \nu}(r_M) = 0$, the mean-value theorem gives some ξ between $\bar{r}_M$ and r_M such that

$$f_{y, \nu}(\bar{r}_M) - f_{y, \nu}(r_M) = f'_{y, \nu}(\xi)(\bar{r}_M - r_M).$$

Therefore,

$$|r_M - \bar{r}_M| = \frac{|f_{y,\nu}(\bar{r}_M)|}{|f'_{y,\nu}(\xi)|} \leq \frac{\delta_M}{a_M c_\psi - L_M}.$$

Taking the supremum gives

$$\|r_M - \bar{r}_M\|_\infty \leq \frac{\delta_M}{a_M c_\psi - L_M}.$$

Since $f'_{y,\nu}(r_M) \neq 0$, the implicit function theorem implies that r_M depends C^1 -smoothly on (y, ν) . The boundary parametrization

$$\Phi_M(y, \nu) = y + r_M(y, \nu)\nu$$

takes values inside the radius- ρ tube. Normal coordinates remain unique there, so Φ_M is injective. It is surjective by construction onto Γ_{M,ε_M} . Thus Γ_{M,ε_M} is a C^1 embedded hypersurface diffeomorphic to $S(N\mathcal{H})$.

Because $f_{y,\nu}$ is strictly decreasing and changes sign from positive to negative, one has

$$q_M(y + s\nu) \geq \varepsilon_M^2 \iff s \leq r_M(y, \nu).$$

Hence

$$S_{M,\varepsilon_M} = \{y + s\nu : 0 \leq s \leq r_M(y, \nu), (y, \nu) \in S(N\mathcal{H})\},$$

which is a normal disk bundle with variable radius and therefore a manifold-with-boundary whose boundary is Γ_{M,ε_M} .

Finally, pair each perturbed boundary point $y + r_M(y, \nu)\nu$ with the reference point $y + \bar{r}_M\nu$. Their distance is $|r_M(y, \nu) - \bar{r}_M|$, so

$$d_H(\Gamma_{M,\varepsilon_M}, \widehat{\Gamma}_{M,\varepsilon_M}) \leq \|r_M - \bar{r}_M\|_\infty \leq \frac{\delta_M}{a_M c_\psi - L_M}.$$

■

Proof of the conditional Fisher-to-geometry proposition.

At the reference point θ_M ,

$$\nabla_{\theta} \eta_M(Z; \theta_M) = \sum_{j \geq 1} \sigma_{M,j} u_{M,j} \varphi_j(Z).$$

Therefore,

$$G_M = \mathbb{E}_\mu \left[\left(\sum_{i \geq 1} \sigma_{M,i} u_{M,i} \varphi_i(Z) \right) \left(\sum_{j \geq 1} \sigma_{M,j} u_{M,j} \varphi_j(Z) \right)^\top \right].$$

Using orthonormality of the φ_j in $L^2(\mu)$ yields

$$G_M = \sum_{j \geq 1} \sigma_{M,j}^2 u_{M,j} u_{M,j}^\top.$$

For the truncation bound,

$$\eta_M - \eta_M^{(k)} = \sum_{j > k} \sigma_{M,j} \langle \Delta\theta, u_{M,j} \rangle \varphi_j + R_M(\Delta\theta; \cdot),$$

so

$$\|\eta_M - \eta_M^{(k)}\|_{C^1} \leq \sum_{j > k} \sigma_{M,j} |\langle \Delta\theta, u_{M,j} \rangle| \|\varphi_j\|_{C^1} + C_{R,M} \|\Delta\theta\|^2.$$

Since $|\langle \Delta\theta, u_{M,j} \rangle| \leq \|\Delta\theta\|$, this becomes

$$\|\eta_M - \eta_M^{(k)}\|_{C^1} \leq \|\Delta\theta\| \sum_{j > k} \sigma_{M,j} B_j + C_{R,M} \|\Delta\theta\|^2.$$

The final geometric displacement bound then follows by applying the perturbative universality theorem to this C^1 -tail estimate. ■

Sampling Corollaries and Fisher Bridge

The theoremic core is deterministic. Sampling appears only when one wants to recover, estimate, or test the geometry from finite data. This separation is deliberate: it prevents recovery theorems from being mistaken for proofs of geometry.

Sampling corollaries.

Once Γ_{M,ε_M} has been shown to be a compact manifold with appropriate regularity and reach control, NSW supplies a high-confidence route to recovering its homology from sufficiently dense i.i.d. samples on or near the manifold; Aamari–Levrard supplies non-asymptotic rates for estimating tangent spaces, curvature, and the manifold itself under quantitative smoothness assumptions. These results are used here only as **black-box lemmas**. They justify what can be inferred once the geometric model is established; they do not establish the geometric model themselves. ⁶

For theorem validation, the correct sampling target is usually the **boundary** Γ_{M,ε_M} , not the full superlevel region. In practice, this can be approximated by band sampling

$$|q_M(x) - \varepsilon_M^2| \leq \delta,$$

inside–outside bracketing followed by bisection along line segments, or normal-probe sampling from an estimated core $\widehat{\mathcal{H}}$. A later empirical paper should state explicitly which sampling regime it uses, because NSW/Aamari–Levrard apply to different observation models.

Conditional Fisher bridge.

Waterfall et al. argue that sloppy nonlinear models commonly exhibit broad log-spaced sensitivity spectra and motivate a Vandermonde ensemble as a universality class. Beckermann–Townsend then proves explicit singular-value decay bounds for displacement-structured matrices, including real Vandermonde matrices.

Those results motivate the Fisher bridge, but they do not imply it. The missing step is exactly the common-spatial-mode separability assumption A_6 , which is why the proposition in this paper is marked conditional and nonstandard. ²

The practical implication is straightforward. If an empirical study finds both rapid Fisher-spectrum decay and evidence that leading spatial perturbation modes are approximately shared across models, then the perturbative theorem converts that spectral decay into a concrete geometric truncation bound. If those shared spatial modes are absent, Fisher decay alone should not be interpreted geometrically.

Statistical Diagnostics and Empirical Protocols

The theorem becomes scientifically meaningful only if its assumptions can be tested. The three central diagnostics are **threshold stability**, **identifiability**, and **modeled perturbation size**.

Threshold stability follows directly from the exact theorem:

$$\left| \frac{d\bar{r}_M}{d\varepsilon_M} \right| = \frac{2\varepsilon_M}{a_M |\psi'(\bar{r}_M)|}.$$

If $|\psi'(\bar{r}_M)|$ is small, the inferred geometry changes rapidly under small perturbations of the threshold. Any empirical paper should therefore report either a threshold sweep or at least local sensitivity bands, rather than treating ε_M as innocuous.

Identifiability is the next issue. The decomposition

$$q_M(y + s\nu) = a_M \psi(s) + b_M + \eta_M(y, \nu, s)$$

is not uniquely parameterized: a_M and ψ admit trivial rescalings. A publishable empirical instantiation should therefore fix a gauge, such as $\psi(0) = 1$ and $\psi'(0) = -1$, or designate one reference model with $a_{M_0} = 1$. The same caution applies to $\mathcal{H}$: nearby low-dimensional sets can sometimes fit the same data almost equally well, so the fitting procedure has to be specified and re-run under bootstrap.

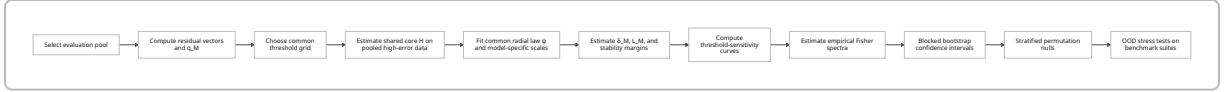
The third diagnostic is the stability margin

$$\mathcal{M}_M := a_M c_\psi - L_M.$$

If $\mathcal{M}_M$ is small or poorly estimated, the perturbative theorem provides weak control. A publication-ready paper should report $\hat{\delta}_M$, $\hat{L}_M$, and $\hat{\mathcal{M}}_M$ with uncertainty intervals, not only point estimates.

The best current public resources suggest a natural validation pipeline. MatPES was designed as a foundational PES dataset emphasizing careful PES sampling and arguing that data quality can outperform sheer scale; official Materials Project documentation describes MatPES as a low-noise, high-coverage dataset for training universal MLIPs and describes MP-ALOE as an active-learning r^2 SCAN dataset exploring under-sampled PES regions. MPtrj is an official Materials Project trajectory-and-single-point resource spanning the database and was used downstream in CHGNet and earlier/later universal MLIP efforts. OMat24 contributes over 110 million DFT calculations and open pretrained models. For evaluation and stress tests, Matbench Discovery, MLIP Arena, LAMBench, MOFSimBench, and phonon benchmarks probe

discovery utility, physics-aware OOD behavior, broad generalization, porous-material reliability, and response-property failures, respectively. 7



The default uncertainty protocol should be a **blocked bootstrap** over materials, trajectory families, or chemically coherent groups, not over individual atomic frames. This is especially important for trajectory-based resources such as MPtrj, whose official description explicitly emphasizes relaxation trajectories and associated ionic-step data. 8

A concrete bootstrap protocol is:

1. Partition the evaluation corpus into blocks $B_1, \dots, B_N$, each block corresponding to one material, one relaxation trajectory, or one chemically coherent group.
2. Resample blocks with replacement to produce B bootstrap datasets.
3. For each replicate, refit $\widehat{\mathcal{H}}, \widehat{\psi}, \widehat{a}_M, \widehat{b}_M, \widehat{\delta}_M$, and $\widehat{L}_M$.
4. Recompute downstream statistics such as boundary displacement, threshold sensitivity, and Fisher-truncation error.
5. Form percentile or BCa confidence intervals.

A concrete stratified permutation protocol is:

1. Define strata by benchmark task, composition family, energy range, and structure family.
2. Within each stratum, permute model labels or residual-field assignments.
3. Recompute the chosen alignment statistic or shared-core fit score.
4. Report the Monte Carlo p -value

$$p = \frac{1 + \#\{T_b \geq T_{\text{obs}}\}}{B + 1}.$$

The most informative empirical quantities are listed below.

Quantity	Estimator	Uncertainty protocol	Failure indicator
Core dimension d	local PCA or manifold estimator on pooled high-error points	blocked bootstrap CI	sharp instability under nearby thresholds
Radial law ψ	monotone regression of q_M on $d_{\widehat{\mathcal{H}}}(x)$	bootstrap band	systematic lack-of-fit
Stability margin $\mathcal{M}_M$	finite-difference derivative along estimated normal directions	blocked bootstrap CI	CI overlaps zero

Quantity	Estimator	Uncertainty protocol	Failure indicator
Boundary displacement	approximate Hausdorff distance to fitted offset	bootstrap CI	displacement comparable to tube radius
Cross-model alignment	principal-angle or shared-core test statistic	stratified permutation p -value	not stronger than shuffled null
Fisher truncation bound	empirical Fisher spectrum plus spatial-mode fit	bootstrap CI	tail bound exceeds stability margin

A strong OOD validation contract is: fit locally on MatPES and MPtrj, scale-test on OMat24, evaluate task utility on Matbench Discovery, and stress the assumptions on MLIP Arena, LAMBench, MOFSimBench, and phonon-centered benchmarks. That benchmark mix reflects the current literature’s strongest evidence for where shared structure might exist and where it might fail. ⁹

Reproducibility of Proofs

The deterministic core of this paper is unusually amenable to formalization because the essential arguments reduce to one-dimensional monotonicity, uniqueness of normal coordinates below reach, and explicit inverse constructions. The parts most suitable for formal proof are the exact theorem, the perturbative root-existence-and-uniqueness lemma, and the Hausdorff displacement bound. The most difficult imported ingredients are the tubular-neighborhood/reach facts, which should be treated as trusted lemmas unless fully formalized separately. Federer’s reach framework and modern expositions of positive reach give the relevant trusted boundary. ¹

A Lean/Mathlib-style theorem skeleton would encode the assumptions as structures and then mechanize the proof steps in the same order used in the paper.

```
-- schematic pseudocode, not executable as-is
structure ErrorGeomData where
  Ω      : Set (EuclideanSpace ℝ (Fin m))
  H      : Set (EuclideanSpace ℝ (Fin m))
  tauH   : ℝ
  rho    : ℝ
  psi    : ℝ → ℝ
  cpsi   : ℝ
  hreach : 0 < tauH
  hrho   : 0 < rho ∧ rho < tauH
  hmono  : ∀ s ∈ Set.Icc 0 rho, deriv psi s ≤ -cpsi

def HighError (q : α → ℝ) (eps : ℝ) : Set α := {x | eps^2 ≤ q x}
def ErrBdry (q : α → ℝ) (eps : ℝ) : Set α := {x | q x = eps^2}
```

```

theorem exact_tubular_universality
(hq :
  ∀ x, x ∈ tube H rho →
    q x = a * psi (dist x H) + b)
(hout :
  ∀ x, x ∉ tube H rho → q x < eps^2)
(hth :
  (eps^2 - b)/a ∈ Set.Ioo (psi rho) (psi 0)) :
∃ rbar,
  ErrBdry q eps = {x | dist x H = rbar} ∧
  HighError q eps = {x | dist x H ≤ rbar} := by
-- monotone inverse + tubular coordinates
sorry

```

A symbolic-verification checklist for the perturbative theorem is similarly clean:

- verify sign change of $f_{y,\nu}(0)$ and $f_{y,\nu}(\rho)$;
- verify strict negativity of $f'_{y,\nu}$;
- apply the implicit function theorem at the unique zero;
- apply the mean-value theorem for the radius perturbation bound;
- pair points along shared normal directions for the Hausdorff estimate.

Because these are low-dimensional analytic steps rather than high-dimensional combinatorial proof-search problems, the core theorem package is a realistic candidate for later full formalization.

Comparison to Recent Studies, Limitations, Counterexamples, and Actionable Revisions

The theorem package here is best understood as a rigorous synthesis of what recent work justifies and what it does not.

Study	Main contribution	What it supports here	What it does not support
Matbench Discovery, 2025	task-aware public evaluation framework and leaderboard for ML energy models	supports public evaluation pools and physically meaningful thresholding	does not prove shared topology or geometry ¹⁰
Systematic softening, 2025	shared PES-softening bias across three uMLIPs caused by near-equilibrium sampling bias	supports the plausibility of some shared error structure	does not imply a tubular universality theorem ¹¹

Study	Main contribution	What it supports here	What it does not support
MLIP Arena, 2025	physics-aware benchmark exposing data leakage, limited transferability, and failure modes	strongly motivates OOD validation of theorem assumptions	does not imply one shared geometry across all domains ¹²
LAMBench, 2026	large-scale benchmark showing a substantial gap to the universal PES	supports a conditional, local theorem rather than sweeping universality	argues against broad universality rhetoric ¹³
MOFSimBench, 2025	difficult porous-material benchmark emphasizing complementary local/global PES probes	supports domain-local testing of the theorem	suggests data quality can dominate architecture identity ¹⁴
Cross-functional transferability, 2025	strong GGA-to- r^2 SCAN energy shifts and poor correlations hinder transfer	shows that $\mathcal{H}$ and ψ may be fidelity-dependent	contradicts naïve multi-fidelity universality ¹⁵
Phonon benchmark, 2025	some models succeed on phonons, others fail despite strong energy/force metrics	supports response-property stress testing of the theorem	warns that one scalarization q_M may not capture all downstream risks ¹⁶

The principal limitations of the theorem are exactly where its strongest assumptions live. The existence of a common core $\mathcal{H}$ is a substantive empirical hypothesis. The common radial profile ψ is also nonstandard and should be interpreted as a local normal form, not a universal physical law. The Fisher bridge requires the strongest assumption of all, namely a common-spatial-mode expansion across models. None of these assumptions should be smuggled in under the label “foundation model.” They should be stated, defended, and stress-tested. The current benchmark ecosystem strongly suggests that they may hold in some benchmark-local regimes and fail in others. ¹⁷

The explicit impossibility result is therefore central.

Counterexample constructing arbitrary hypersurfaces as regular level sets.

Let $\Sigma \subset \Omega$ be any compact embedded C^∞ hypersurface. Then there exists a smooth function $f : \Omega \rightarrow \mathbb{R}$ such that 0 is a regular value of f and

$$f^{-1}(0) = \Sigma.$$

Consequently, given two non-diffeomorphic compact embedded hypersurfaces $\Sigma_1, \Sigma_2 \subset \Omega$, there exist smooth scalarized error fields q_1, q_2 and thresholds c_1, c_2 such that

$$q_i^{-1}(c_i) = \Sigma_i, \quad i = 1, 2.$$

Hence no unrestricted universality theorem over all smooth error fields can hold.

Proof.

By the tubular neighborhood theorem, there exists $r > 0$ and a diffeomorphism

$$\Phi : \Sigma \times (-r, r) \rightarrow U \subset \Omega$$

onto a tubular neighborhood of Σ . Define the local signed normal-coordinate function by

$$f_0(\Phi(y, t)) := t.$$

Then $f_0^{-1}(0) = \Sigma \cap U$, and 0 is a regular value because $\partial_t f_0 = 1$ along Σ .

Choose a smooth odd saturation map $\chi : (-r, r) \rightarrow \mathbb{R}$ such that $\chi(t) = t$ for $|t| \leq r/3$ and $\chi(t) = \pm r/2$ for $\pm t \geq 2r/3$. Then $g_0 := \chi \circ f_0$ equals f_0 near Σ and is locally constant near ∂U . Extend g_0 smoothly to all of Ω by keeping those constants on the complementary components of $\Omega \setminus U$. The resulting smooth $f : \Omega \rightarrow \mathbb{R}$ satisfies $f^{-1}(0) = \Sigma$, and 0 remains a regular value because $f = f_0$ near Σ . Applying this construction to two non-diffeomorphic hypersurfaces yields the claim. The required tubular-neighborhood machinery is standard for compact smooth embedded hypersurfaces. ¹⁸ ■

The publication path for a theorem paper based on this framework is short and concrete.

Priority	Actionable revision	Why it is essential
Highest	Define the error object explicitly as $\Gamma_{M,\varepsilon}$, $S_{M,\varepsilon}$, $\mathcal{H}$, or a stratified/measure-theoretic alternative	prevents dimension and topology confusion
Highest	Make the exact tubular universality theorem the headline theorem	it is the strongest self-contained result proved here
Highest	State the counterexample prominently	shows why the assumptions are necessary
High	Flag A_2 and A_6 explicitly as nonstandard	prevents hidden-assumption universality
High	Keep Fisher claims conditional unless common spatial modes are empirically verified	Waterfall/Beckermann motivate, but do not prove, the bridge ²
High	Validate first on MatPES/MPtrj and then stress-test on Matbench Discovery, MLIP Arena, LAMBench, MOFSimBench, OMat24, and cross-functional settings	aligns theorem testing with the strongest current benchmarks ¹⁹
Medium	Fix a gauge for (a_M, ψ, b_M) and report threshold sensitivity	ensures interpretability
Medium	Add response-property and OOD stress tests, not just static residual fits	recent phonon and benchmark work shows static metrics are insufficient ²⁰

Reproducible Reporting Checklist and Conclusion

A publication-ready theorem paper should satisfy a reproducibility standard at least as strict as the current public benchmark ecosystem.

Item	What must be reported
Configuration domain	exact definition of Ω , including symmetry reduction and whether composition/atom count are fixed
Residual vector	full formula for r_M , units, and weights w_E, w_F, w_σ
Scalarized error	exact definition of q_M and any normalization
Error object	whether the theorem concerns $\Gamma_{M,\varepsilon}, S_{M,\varepsilon}, \mathcal{H}$, or another object
Assumptions ledger	every assumption, with standard versus nonstandard status and point of use
Dependency map	which results are proved in-paper and which are black-box imports
Gauge choice	how the (a_M, ψ, b_M) non-identifiability is fixed
Threshold protocol	common threshold grid or physically justified thresholds, plus sensitivity sweep
Stability diagnostics	estimates and CIs for $\delta_M, L_M, a_M c_\psi - L_M$, and $d\bar{r}_M/d\varepsilon_M$
Sampling design	how near-boundary samples are formed and which recovery theorem is invoked
Bootstrap protocol	resampling unit, replicate count, CI construction, and random seed policy
Permutation protocol	strata, test statistic, and Monte Carlo p -value formula
Benchmark matrix	exact tasks and versions for MatPES, MPtrj, Matbench Discovery, MLIP Arena, LAMBench, MOFSimBench, OMat24, and any cross-functional splits
Code availability	scripts for fitting $\widehat{\mathcal{H}}, \widehat{\psi}$, Fisher spectra, and uncertainty procedures
Formal artifacts	theorem statements plus Lean/Mathlib-oriented scripts or pseudocode
Failure cases	explicit domains or thresholds where assumptions fail and no theorem is claimed

The main conclusion is precise. A **single-source, publish-ready universality theorem** for MLIP error geometry is possible, but only in a conditional and mathematically explicit sense. The exact and perturbative tubular theorems proved here give a rigorous universality result for a class F defined by a shared low-dimensional error core and a common radial error law, with explicit quantitative stability under model-specific perturbations. The counterexample shows why nothing stronger can hold without extra structure. The Fisher bridge is worth including only as a conditional extension under a clearly labeled nonstandard common-spatial-mode assumption. NSW and Aamari–Levrard then supply sampling

corollaries once the geometric model is already justified. This is, at present, the strongest theorem package that is both mathematically defensible and scientifically aligned with the benchmark evidence from Matbench Discovery, MLIP Arena, LAMBench, MOFSimBench, MatPES, MPtrj, and OMat24. ²¹

If a specific target paper is later provided, this theorem package can be re-instantiated against its exact definitions, datasets, benchmark scope, and claimed universality class without changing the logical backbone of the derivation.

¹ https://users.math.msu.edu/users/iwenmark/Teaching/MTH995/Papers/MMod_Federer.pdf

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² <https://arxiv.org/abs/cond-mat/0605387>

<https://arxiv.org/abs/cond-mat/0605387>

³ ⁵ ¹⁰ <https://www.nature.com/articles/s42256-025-01055-1>

<https://www.nature.com/articles/s42256-025-01055-1>

⁴ ⁶ ²¹ <https://link.springer.com/article/10.1007/s00454-008-9053-2>

<https://link.springer.com/article/10.1007/s00454-008-9053-2>

⁷ ⁹ ¹⁹ <https://arxiv.org/abs/2503.04070>

<https://arxiv.org/abs/2503.04070>

⁸ <https://docs.materialsproject.org/services/ml-and-ai-applications/mptrj>

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¹¹ <https://www.nature.com/articles/s41524-024-01500-6>

<https://www.nature.com/articles/s41524-024-01500-6>

¹² <https://arxiv.org/abs/2509.20630>

<https://arxiv.org/abs/2509.20630>

¹³ ¹⁷ <https://www.nature.com/articles/s41524-025-01929-3>

<https://www.nature.com/articles/s41524-025-01929-3>

¹⁴ <https://www.nature.com/articles/s41524-025-01872-3>

<https://www.nature.com/articles/s41524-025-01872-3>

¹⁵ <https://www.nature.com/articles/s41524-025-01796-y>

<https://www.nature.com/articles/s41524-025-01796-y>

¹⁶ ²⁰ <https://www.nature.com/articles/s41524-025-01650-1>

<https://www.nature.com/articles/s41524-025-01650-1>

¹⁸ <https://geometrica.saclay.inria.fr/team/Fred.Chazal/papers/ccslmt-icutn-17/inference.pdf>

<https://geometrica.saclay.inria.fr/team/Fred.Chazal/papers/ccslmt-icutn-17/inference.pdf>